BHUVANESAN N

Software Developer

Java | Spring Boot | MySQL | REST APIs

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PROFILE SUMMARY

Computer Science undergraduate with hands-on experience in Java, Spring Boot, REST APIs, SQL, and MongoDB, gained through academic and personal software development projects. Strong foundation in Object-Oriented Programming, Data Structures & Algorithms, and backend application development, with 170+ LeetCode problems solved. Experienced in building scalable CRUD applications, integrating databases, and developing IoT-enabled solutions. Passionate about designing efficient backend systems, writing clean, maintainable code, and continuously learning modern software engineering technologies.

PROJECTS

Digital AI Twin IoT · Python · AI/ML

- Developed a Smart Energy Meter using ESP32 and LCD to monitor and display real-time current, power, and energy consumption.
- Processed sensor inputs to calculate electrical parameters and present live readings on the embedded display.
- Implemented real-time energy monitoring to demonstrate efficient power usage aligned with Sustainable Development Goal (SDG 7).
- Designed an embedded solution capable of continuous energy usage tracking for smart monitoring applications.

Smart Energy Meter ESP32 · LCD · Embedded Systems · IoT

- Developed a Smart Energy Meter using ESP32 and LCD to monitor and display real-time current, power, and energy consumption.
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Employee Management System HTML · CSS · SQL

- Developed an Employee Management System with a responsive web interface for managing employee records.
- Implemented CRUD operations to add, update, delete, and retrieve employee information using SQL.
- Designed structured input forms for efficient employee data management and validation.
- Built a user-friendly interface to simplify employee record management and improve usability.

Micro Failure Whisperer (Ongoing) Python · Machine Learning · Predictive Maintenance

- Developing an AI-powered predictive maintenance system to detect micro-level machine failures using sensor and vibration data.
- Applying anomaly detection techniques to identify early signs of equipment degradation before breakdowns occur.
- Designing a real-time monitoring pipeline to process sensor data and generate predictive maintenance alerts.
- Building a scalable architecture capable of supporting industrial machine health monitoring applications.

Digital Smell (Ongoing) IoT · Gas Sensors · Embedded AI

- Developing an IoT-based gas leakage detection system using high-sensitivity gas sensors and embedded AI.
- Implementing real-time monitoring and anomaly detection to identify minute gas leakages accurately.
- Designing an intelligent alert system to improve industrial safety through early hazard detection.
- Building an embedded solution for continuous environmental monitoring in safety-critical applications.

ScriptGuard AI (Ongoing) Python · Computer Vision · OCR

- Developing an intelligent paper-scanning mechanism for automated document digitization and verification.
- Integrating OCR and image-processing techniques to extract, validate, and organize scanned content.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++
Frontend & Web Technologies: HTML5, CSS3, JavaScript, Bootstrap, Responsive Web Design
Backend Technologies: Core Java, Spring, Spring Boot, Spring MVC, REST APIs, JDBC, Hibernate, JPA, CRUD Operations
Databases: MySQL, MongoDB
Tools & Platforms: Git, GitHub, Maven, Postman, IntelliJ IDEA, Eclipse, VS Code
Core Concepts: OOP, Data Structures & Algorithms, DOM Manipulation, Anomaly Detection
Cloud & Data: Snowflake
IoT & Embedded: ESP32, Sensor Integration, Embedded AI, Gas & Vibration Sensors

INTERPERSONAL SKILLS

- Communication
- Presentation
- Problem-Solving
- Team Player
- Time Management
- Attention to Detail

• Adaptability

WORKSHOPS & HACKATHONS

HACKXTREME'26 — 24-Hour National-Level Hackathon, K. Ramakrishnan College of Engineering, Trichy

- Built an AI-powered Digital Twin model simulating urban infrastructure from IoT data for real-time prediction of traffic congestion, energy demand, and pollution levels.
- Recognized by Media Wave for technical performance and shortlisted for an internship opportunity.

Smart India Hackathon (SIH)

- Designed an automated pesticide sprinkler system using logic-based control to improve agricultural efficiency.
- Developed a prototype focused on reducing manual intervention while optimizing pesticide spraying.
- Collaborated with team members to design an innovative solution addressing real-world agricultural challenges.

MSME

- Presented Micro Failure Whisperer and Digital Smell as AI-driven industrial safety and predictive maintenance solutions.
- Demonstrated the application of machine learning and IoT technologies for early fault detection and industrial monitoring.

Snowflake Workshop — 1 Week

- Completed hands-on training covering Snowflake architecture, cloud data warehousing, and SQL-based data management.
- Gained practical exposure to data loading, querying, and cloud-based analytics workflows.

IoT Workshop — 1 Month

- Completed practical training in ESP32 programming, sensor integration, and embedded system development.
- Built real-time IoT applications involving sensor data acquisition and hardware-software communication.

CERTIFICATIONS

- AI/ML (Dreams AI)
- SQL (Oracle)
- Java (Oracle)
- MongoDB (ICT Academy)
- Figma (Simplelearn)
- Python / C & C++ (Kaggle/mygreatlearning)

EDUCATION

B.E. Computer Science and Engineering (CSE) — PSNA College of Engineering and Technology, Dindigul — 2024 – 2028 | CGPA: 8.6

XII — St. John's Matriculation Higher Secondary School, Madurai — May 2024 | 93%

X — The Nallamani Matriculation Higher Secondary School, Madurai — July 2022 | 92%

ACHIEVEMENTS

- Recognized by Media Wave at HACKXTREME'26 (24-Hour National-Level Hackathon) for technical performance and shortlisted for an internship opportunity.
- Solved 170+ problems on LeetCode across Arrays, Strings, Mathematics, and core DSA topics.
- Won mathematics competitions, demonstrating strong analytical thinking and problem-solving abilities.